Week 1 Module

C and C++ Programming

C Programming

① C Syntax and Basic Structure

C is a procedural and structured programming language. Every standard C program has a precise structural layout.

Header files (#include <stdio.h>)

This pre-processor directive includes the standard Input/output library. It allows the program to use the core functions, such as printf() for output and scanf() for input.

Main function (int main())

This is the ~~mad~~ mandatory entry point of any C program. Execution starts at the opening brace { of main(). Execution ends at the terminating semicolon of return 0;.

Output function (printf)

Unlike C++, C uses printf() function to display text on the screen. Double quotes let you pass format specifiers or escape sequences like \n (newline).

Semicolon (;): Every statement in C must end with a semicolon. It is a statement terminator for the compiler.

② Conditions and loops

Conditional statements (if-else): used to make decisions. The if block runs when the condition is true. If it is false, the program skips it or executes the optional else block.

loops (for loop): To repeat a block of code a certain number of times. It is composed of intialization, condition checking and increment / decrement expressions.

Nested loops - for star patterns to print two dimensional grid patterns we need a loop in another loop.

Outer loop: Manages the rows (vertical progression)

Inner loop: Handles the columns (prints spaces or stars horizontally for the current row).

DELTA
Date / /
Pg No.

(3) Programming problems

Solve the given star pattern Questions

(1) square

O/P :-
```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

```
#include <stdio.h>
int main (){
   int n=5; // total rows and columns
   // outer loop for rows
   for (int i=1; i<=n; i++){
   // Inner loop for columns
   for (int j=1; j<=n; j++){
       printf("*");
   }
   printf ("\n"); // move to the next line
   }
   return 0;
}
```

② Hollow square pattern

```c
#include <stdio.h>
int main(){
int n=5;
for(int i=1; i<=n; i++){
for(int j=1; j<=n; j++){
// Print star only at the boundaries
if(i==1 || i==n || j==1 || j==n){
printf("*");
} else {
printf(" "); // print empty space inside
}}
printf("\n");
}
return 0;
}
```

O/P:—

```
*****
*   *
*   *
*   *
*****
```

3) Triangle:

```
#include <stdio.h>
int main(){
int n=5;
for (int i=1; i<=n; i++){
// Inner loop runs up to the current row number 'i'
for (int j=1; j<=i; j++){
printf("*");
}
printf("\n");
}
return 0;
}
```

O/P:-

```
*
* *
* * *
* * * *
* * * * *
```

(4) Inverted Triangle

```
#include <stdio.h>
int main (){
int n = 5;
//outer loop Counts down from 'n' to 1
for (int i = n; i >= 1; i--){
for (int j = 1; j <= i; j++){
printf("*");
}
printf("\n");
}
return 0;
}
```

O/P :-

```
* * * * *
* * * *
* * *
* *
*
```

(5) Pyramid

```
#include <stdio.h>
int main (){
int n=5;
for(int i = 1; i<=n; i++) {
// print leading spaces
for (int j = 1; j<=n-i; j++){
printf(" ");
}
// print stars based on odd numbers formula.
for (int k = 1; k <=(2*i-1); k++){
printf("*");
}
printf("\n");
}
return 0;
}
```

O/p:—

```
    *
   * * *
  * * * * *
 * * * * * * *
* * * * * * * * *
```

(6) Inverted pyramid

```c
#include <stdio.h>
int main(){
int n=5;
// outer loop handles rows in reverse order
for (int i=n; i>=1; i--){
// print spaces
for(int j=1; j<=n-i; j++){
printf(" ");
}
// print stars
for(int k=1; k<=(2*i-1); k++){
printf("*");
}
printf("\n");
}
return 0;
}
```

O/p:-

```
*********
 *******
  *****
   ***
    *
```

(7) Half diamond

```
#include <stdio.h>
int main (){
int n = 5;
// upper half: Growing Triangle.
for( int i =1; i <= n; i++){
for( int j = 1; j <= i; j++) {
printf ("* ");
}
printf( "\n");
}
// lower Half; shrinking Inverted Triangle
for (int i = n -1; i >= 1; i--) {
for (int j = 1; j <= i; j++){
printf("* ");
}
printf("\n");
}
return 0;
}
```

O/P:-

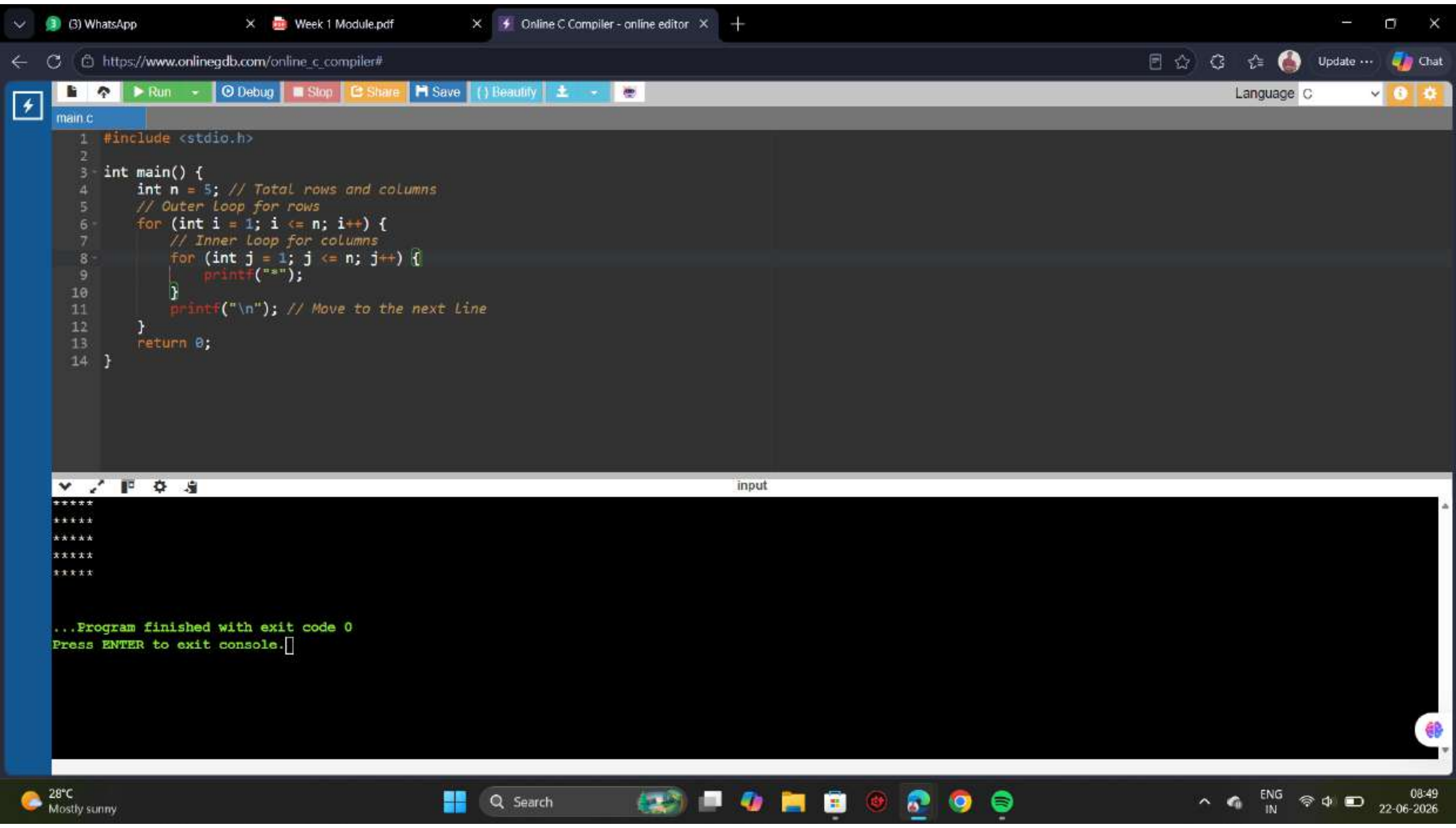
```
*
* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
*
```

(8) Diamond

```
#include <stdio.h>
int main (){
int n=5;
// upper Half: Regular pyramid
for (int i=1; i<=n; i++){
for (int j=1; j<=n-i; j++){
printf(" ");
for (int k=1; k<=(2*i-1); k++)
printf("*");
printf("\n");
}
// lower Half: Inverted pyramid
for (int i=n-1; i>=1; i--){
for (int j=1; j<=n-i; j++)
printf(" ");
for(int k=1; k<=(2*i-1); k++)
printf("*");
printf("\n");
}
return 0;
}
```

O/P:-

```
    *
   ***
  *****
 *******
  *****
   ****
    *
```


(3) WhatsApp
Week 1 Module.pdf
Online C Compiler - online editor
https://www.onlinegdb.com/online_c_compiler#
Run
Debug
Stop
Share
Save
{ } Beautify
Language C
main.c
#include <stdio.h>
int main() {
int n = 5; // Total rows and columns
// Outer loop for rows
for (int i = 1; i <= n; i++) {
// Inner loop for columns
for (int j = 1; j <= n; j++) {
printf("*");
}
printf("\n"); // Move to the next line
}
return 0;
}
input
*****
*****
*****
*****
*****
...Program finished with exit code 0
Press ENTER to exit console.
28°C
Mostly sunny
Search
ENG
IN
08:49
22-06-2026

main.c

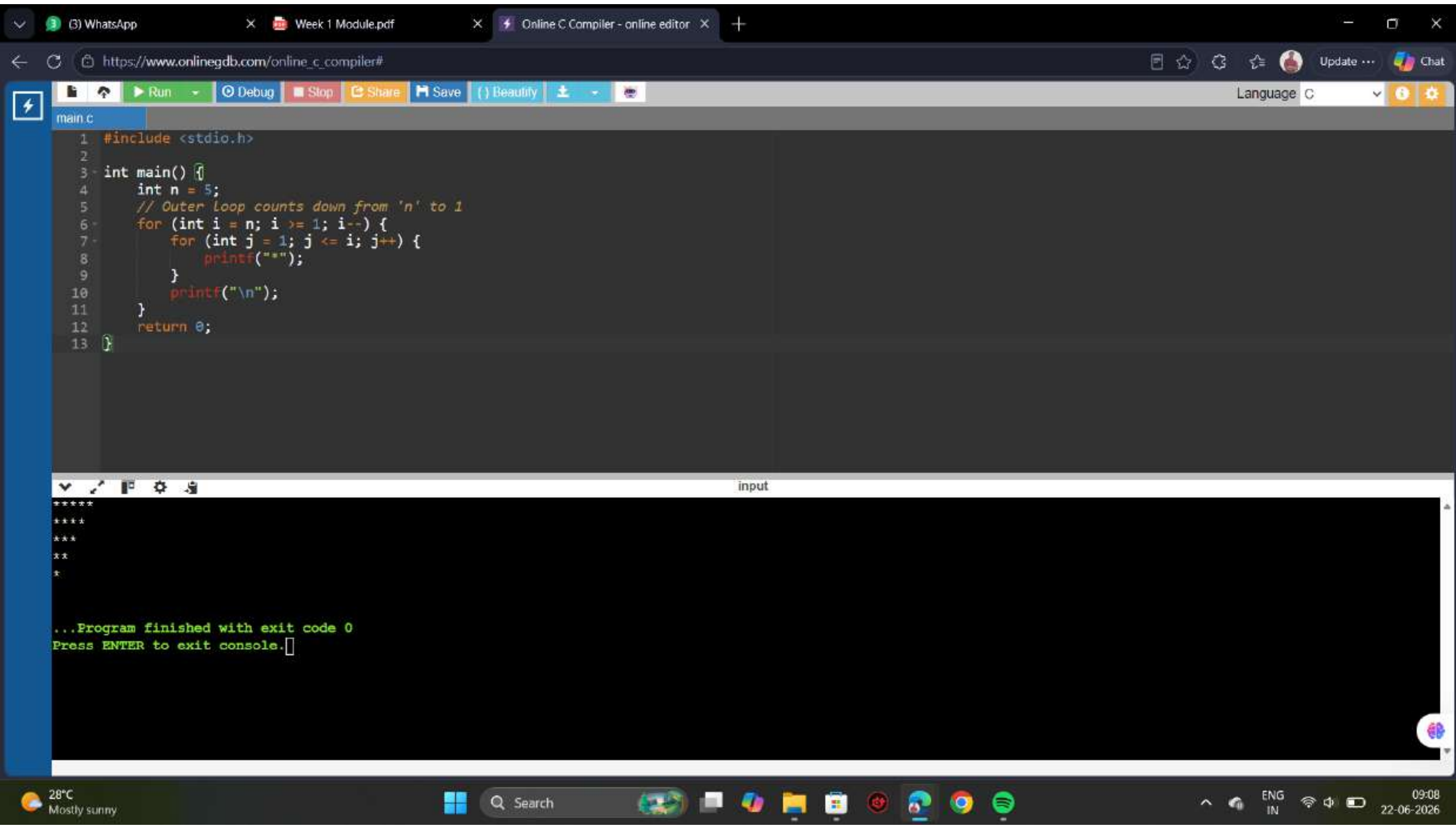
```c
#include <stdio.h>

int main() {
    int n = 5;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) {
            // Print star only at the boundaries
            if (i == 1 || i == n || j == 1 || j == n) {
                printf("*");
            } else {
                printf(" "); // Print empty space inside
            }
        }
        printf("\n");
    }
    return 0;
}
```

input

```
*****
*   *
*   *
*   *
*****


...Program finished with exit code 0
Press ENTER to exit console.
```

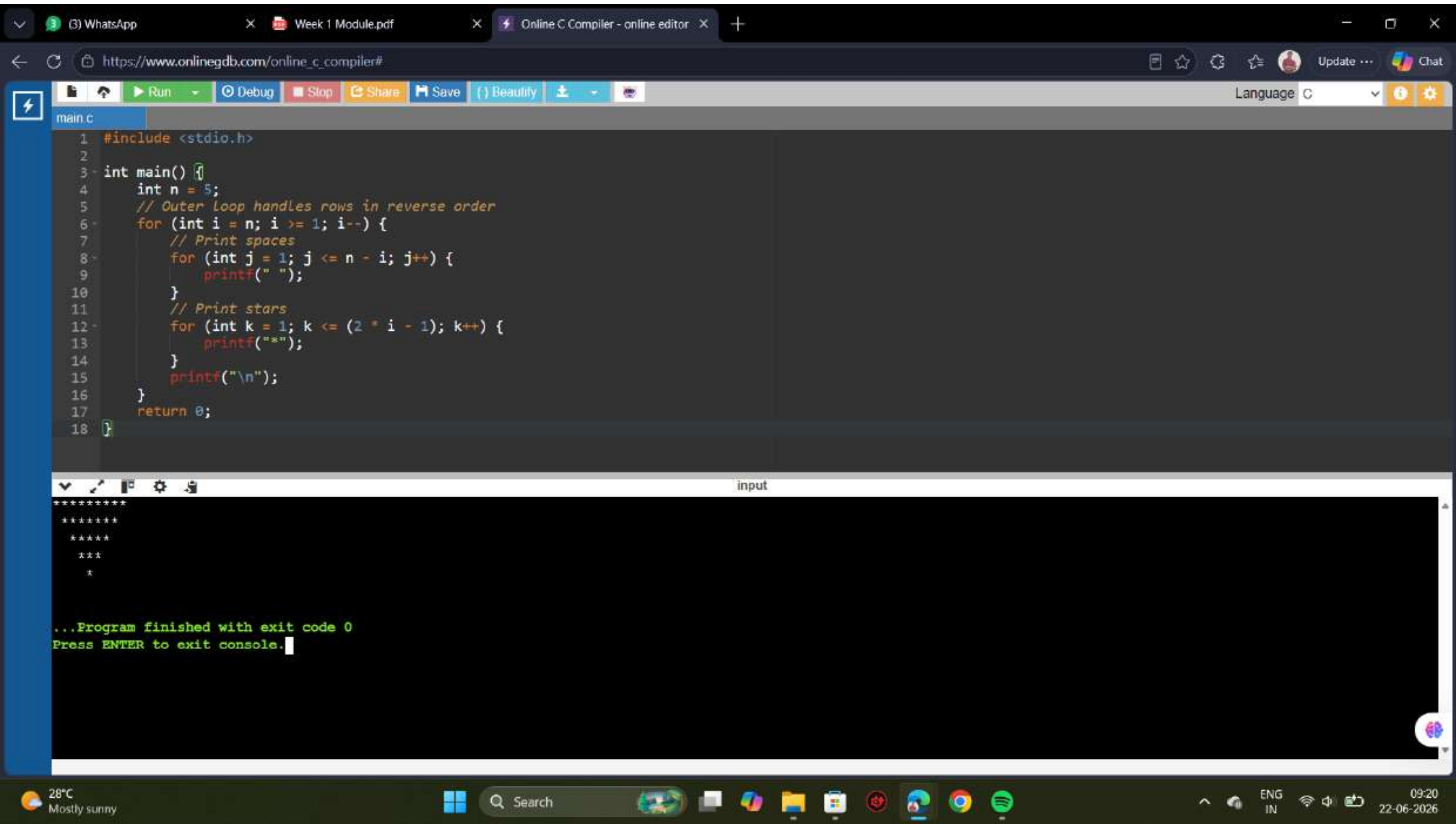
```c
#include <stdio.h>

int main() {
    int n = 5;
    for (int i = 1; i <= n; i++) {
        // Inner loop runs up to the current row number 'i'
        for (int j = 1; j <= i; j++) {
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```

```
*
**
***
****
*****


...Program finished with exit code 0
Press ENTER to exit console.
```


(3) WhatsApp
Week 1 Module.pdf
Online C Compiler - online editor
https://www.onlinegdb.com/online_c_compiler#
Update
Chat
Run
Debug
Stop
Share
Save
{ } Beautify
Language
C
main.c
#include <stdio.h>
int main() {
int n = 5;
// Outer loop counts down from 'n' to 1
for (int i = n; i >= 1; i--) {
for (int j = 1; j <= i; j++) {
printf("*");
}
printf("\n");
}
return 0;
}
input
*****
****
***
**
*
...Program finished with exit code 0
Press ENTER to exit console.
28°C
Mostly sunny
Search
ENG
IN
09:08
22-06-2026

```c
#include <stdio.h>

int main() {
    int n = 5;
    for (int i = 1; i <= n; i++) {
        // Print leading spaces
        for (int j = 1; j <= n - i; j++) {
            printf(" ");
        }
        // Print stars based on odd number formula: 2 * i - 1
        for (int k = 1; k <= (2 * i - 1); k++) {
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```

```
    *
   ***
  *****
 *******
*********


...Program finished with exit code 0
Press ENTER to exit console.
```


(3) WhatsApp
Week 1 Module.pdf
Online C Compiler - online editor
https://www.onlinegdb.com/online_c_compiler#
Run
Debug
Stop
Share
Save
{ } Beautify
Language C
main.c
#include <stdio.h>
int main() {
int n = 5;
// Outer loop handles rows in reverse order
for (int i = n; i >= 1; i--) {
// Print spaces
for (int j = 1; j <= n - i; j++) {
printf(" ");
}
// Print stars
for (int k = 1; k <= (2 * i - 1); k++) {
printf("*");
}
printf("\n");
}
return 0;
}
input
****
****
*****
***
*
...Program finished with exit code 0
Press ENTER to exit console.
28°C
Mostly sunny
Search
ENG
IN
09:20
22-06-2026

```c
#include <stdio.h>

int main() {
    int n = 5;
    // Upper Half: Growing Triangle
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) {
            printf("*");
        }
        printf("\n");
    }
    // Lower Half: Shrinking Inverted Triangle
    for (int i = n - 1; i >= 1; i--) {
        for (int j = 1; j <= i; j++) {
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```

```
*
**
***
****
*****
****
***
**
*


...Program finished with exit code 0
Press ENTER to exit console.
```

main.c

```c
#include <stdio.h>

int main() {
    int n = 5;
    // Upper Half: Regular Pyramid
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n - i; j++) printf(" ");
        for (int k = 1; k <= (2 * i - 1); k++) printf("*");
        printf("\n");
    }
    // Lower Half: Inverted Pyramid
    for (int i = n - 1; i >= 1; i--) {
        for (int j = 1; j <= n - i; j++) printf(" ");
        for (int k = 1; k <= (2 * i - 1); k++) printf("*");
        printf("\n");
    }
    return 0;
}
```

```
    *
   ***
  *****
 *******
*********
 *******
  *****
   ***
    *


...Program finished with exit code 0
Press ENTER to exit console.
```